AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 12

Lab Title Sound Signals

Student Name: Ali Abbas Khan

Reg. No: 230661

Objective: _____

_____

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 12

Title: Sound Signal

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. Maryam
Date**:** December 17, 2025

Lab Task:

1. You Are Given with the Audio, Read And write In MATLAB And Plot the audio Signal.

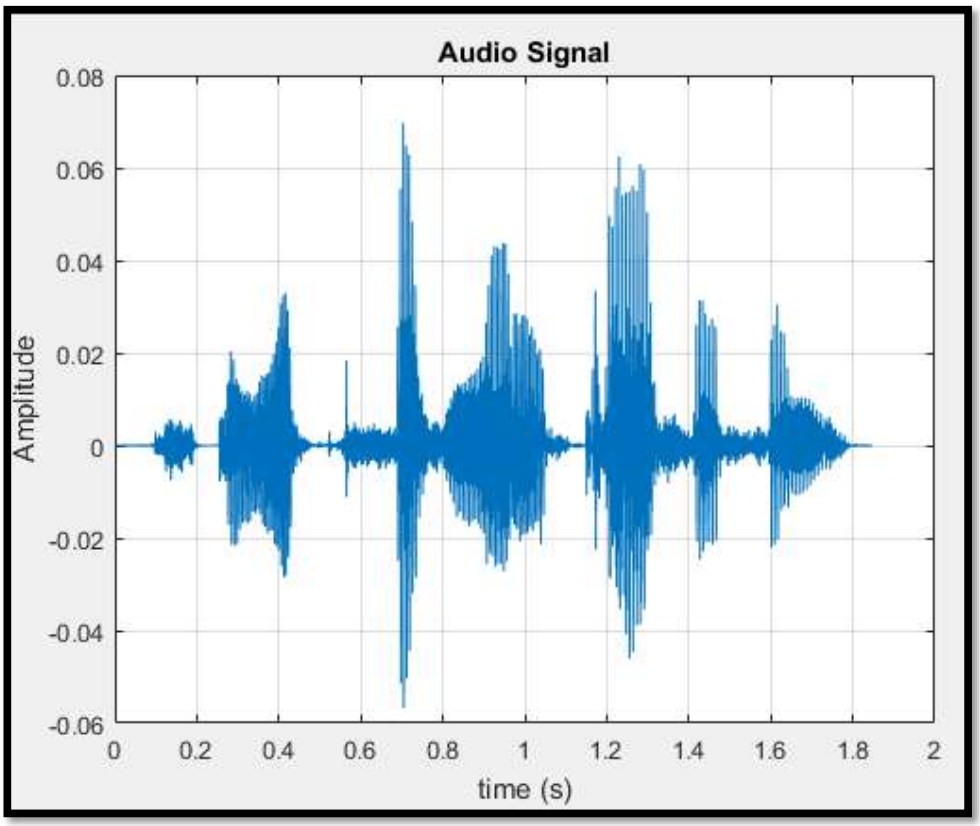
Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L12\LabTask.m

LabTask.m | LabTask2.m

```
1 -    [y,Fs]=audioread('Lab12.wav');
2 -    sound(y,Fs);
3 -    t=(0:length(y)-1)/Fs;
4 -    plot(t,y);
5 -    xlabel('time (s)');
6 -    ylabel('Amplitude');
7 -    title('Audio Signal');
8 -    grid on;
```

Output:



2. Download the Audio From Google Read and write the Audio in Matlab and decrease its Playback speed to 0.5x.

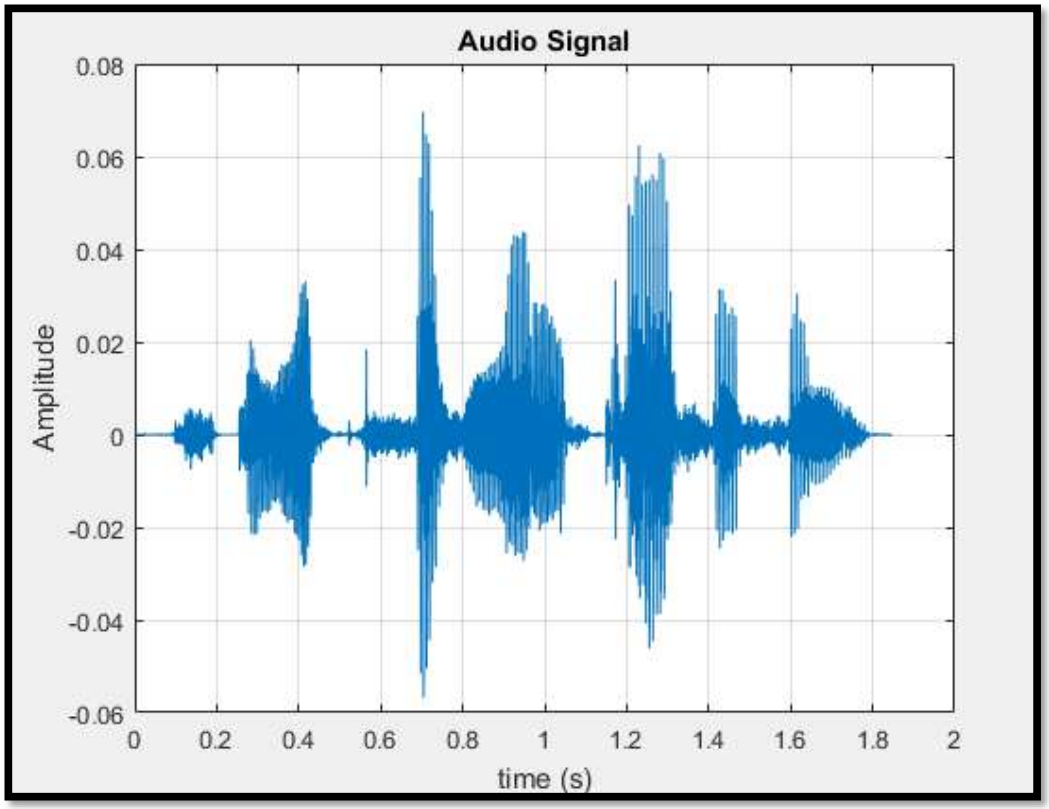
Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L12\LabTask2.m

LabTask.m | LabTask2.m

```
1 -    [y2,Fs]=audioread('Lab12.wav');
2 -    audio=y2;
3 -    new_Fs=Fs*0.5;
4 -    audiowrite('Lab12_new.wav',audio,new_Fs);
5 -    [y,Fs2]=audioread('Lab12_new.wav');
6 -    sound(y,Fs2);
7 -    t=(0:length(y)-1)/Fs;
8 -    plot(t,y);
9 -    xlabel('time (s)');
10 -   ylabel('Amplitude');
11 -   title('Audio Signal');
12 -   grid on;
```

Output:



Conclusion:

The lab demonstrated that an audio signal is essentially a discrete set of numbers stored at a specific frequency. Altering that frequency changes our perception of the sound without changing the data itself, while plotting provides a visual insight into the signal's envelope and characteristics.